

Karthikeyan Sugavanan

Boston, MA 02120 | (857) 506-4888 | [EMAIL](#) | [GitHub](#) | [LinkedIn](#)

Data Science Graduate Student | AI Research Engineer | Quantitative Analytics Specialist

Available: January 2025 - May 2026

EDUCATION

Northeastern University

Master of Science in Data Science (GPA: 3.85/4.0)

Coursework: Deep Learning, Transformer Models, Supervised/Unsupervised Machine Learning, Algorithms, LLM, NLP, Cloud Technologies

Vellore Institute of Technology

Bachelor of Technology in Computer Science (GPA: 3.4/4.0)

Focus: AI, Computer Vision, NLP, Algorithmic Trading, Database Systems, High-Performance Computing, Data Mining

Boston, MA

Jan. 2024 – May 2026

Chennai, India

Aug. 2019 – May 2023

PROFESSIONAL EXPERIENCE

Data Science Intern

Jan 2023 – Aug 2023

VIT Chennai, India

- Engineered **AutoML pipeline** processing **15K+** applications with **ensemble methods**, **GridSearchCV optimization**, and **Scikit-learn frameworks**, achieving **25% accuracy improvement** and **40% latency reduction**
- Developed **transformer NLP models** using **BERT embeddings**, **TF-IDF vectorization**, **word2vec**, **spaCy preprocessing**, and **dimensionality reduction (PCA, t-SNE)** for admission prediction
- Implemented **weighted scoring algorithm** with **ensemble voting classifiers (Random Forest, XGBoost)**, reducing time by **60%**
- Deployed **FastAPI** with **Redis caching**, **PostgreSQL**, and **containerized microservices** for real-time inference and automation

Data Science Research Intern

Jan 2022 – Dec 2022

VIT Business School, Chennai, India

- Led **12-month research program** standardizing **700K+** records across **6-year dataset**, resulting in **IEEE publications**
- Architected **ETL infrastructure** using **Apache Kafka**, **KNIME**, **SQL procedures**, and **Delta Lake** with **ACID transactions**
- Applied **ensemble ML models (Random Forest, SVM-RBF, XGBoost)** with **GridSearchCV** and **neural networks**
- Deployed **15+ Tableau dashboards** with **real-time connections** for **2 departments** and **5+ faculty stakeholders**

PROJECTS AND RESEARCH

Algorithmic Trading System with Stochastic Deep Learning | QuantFinance, Deep Learning

- Achieved **97% directional accuracy** and **15% performance improvement** through **hybrid CNN-LSTM architecture** with advanced quantitative pattern recognition
- Engineered **multi-model ensemble** combining **LSTM temporal networks**, **CNN pattern detection**, and **XGBoost gradient boosting** for time-series forecasting
- Implemented **comprehensive technical indicators (RSI, MACD, Bollinger Bands)** with **stochastic optimization algorithms** and **Monte Carlo simulations**
- Deployed **Streamlit interface** with **real-time backtesting framework**, **12-month validation**, and **risk metrics**

RAG-Enhanced Knowledge System with RARE Framework | NLP, GenAI

- Achieved **42% hallucination reduction** and **22% accuracy improvement** through **RARE framework**
- Architected **hybrid retrieval system** with **FAISS vector indexing**, **DuckDB SQL filtering**, **cross-encoder reranking**, and **hierarchical document chunking**
- Integrated **LLaMA-2 7B**, **GPT-3.5**, and **Zephyr-7B-Alpha** models with **BAAI/bge-base-en embeddings** processing documents.
- Implemented **multi-step reasoning pipeline** with **confidence estimation**, **contradiction resolution**, and **29x cost efficiency**

Enterprise Market Intelligence Platform with Advanced Analytics | Data Engineering, NLP

- Improved **20% positioning strategy** and **12% intelligence accuracy** increase through **automated multi-platform data collection**
- Orchestrated **enterprise-grade scraping infrastructure** from **50+ e-commerce platforms** using **Selenium WebDriver**, **BeautifulSoup**, **Scrapy**, and **rotating proxy servers**
- Applied **comprehensive advanced NLP pipeline** including **VADER sentiment analysis**, **BERT-based competitive intelligence extraction**, **TF-IDF vectorization** and **predictive market trend forecasting** with **time-series analytics** and **machine learning algorithms**
- Deployed **real-time Power BI dashboards** with **PostgreSQL OLTP backend**, **Redis distributed caching layer**, **Apache Airflow DAG orchestration**, **automated ETL pipelines**, **API integrations**, and **25K+ customer reviews sentiment analysis**

AI-Powered Credit Risk Assessment with Explainable Analytics | Risk Analytics, XAI

- Achieved **94.95% AUC classification accuracy** with **50% manual review reduction** through **ensemble gradient boosting models**
- Engineered **200+ sophisticated behavioral features** using **advanced customer segmentation algorithms**, **temporal aggregation techniques**, **payment pattern analysis**, **risk profiling algorithms**, **feature selection methodologies**, **correlation analysis**, and **statistical significance testing**
- Implemented **production-ready ensemble models** combining **XGBoost gradient boosting**, **LightGBM efficient training**, **CatBoost categorical handling** with **GridSearchCV hyperparameter optimization**, **cross-validation frameworks**, **model stacking techniques**, and **performance evaluation metrics**
- Generated **\$4.5M+ projected ROI** through **SHAP explainability**, **real-time scoring APIs**, **model monitoring**, and **dashboards**

Real-Time Audio Threat Detection System | Deep Learning, IoT

- Achieved **96.6% classification accuracy** with **2.3-second response time** for **IEEE Access published emergency threat detection system** using **real-time audio processing**, **pattern recognition algorithms**, and **optimized inference pipelines**
- Built **sophisticated multi-model deep learning architecture** using **bidirectional LSTM network** etc with **TensorFlow/Keras frameworks** and **GPU acceleration**
- Preprocessed **15K+ samples** using **librosa feature extraction**, **Mel-spectrogram generation**, **FFT analysis** and **data augmentation**
- Integrated **comprehensive multi-channel alerting system** with **Twilio SMS APIs**, **SMTP email services**, **Docker containerization**, **Kubernetes orchestration**, and **cloud microservices deployment**

TECHNICAL EXPERTISE

Languages: Python, SQL, C++, R, Java, Scala, JavaScript | **ML:** PyTorch, TensorFlow, Scikit-learn, XGBoost, LightGBM, CatBoost

Data Engineering: Spark, Kafka, Flink, dbt, Delta Lake, KNIME | **Cloud:** AWS, GCP, Azure

DB: PostgreSQL, MongoDB, Redis, DuckDB | **Visualization:** Tableau, Power BI, Plotly, Matplotlib, Streamlit | **DevOps:** Docker, Kubernetes, Git, Terraform, Jupyter, Hadoop

Specialized: NLP (NLTK, spaCy, Transformers), Time-Series, Quantitative Finance, Graph Neural Networks, Reinforcement Learning

Analytics: A/B Testing, Causal Inference, Bayesian Statistics, Monte Carlo Methods, Statistical Modeling, Feature Engineering